

Thermal sensors integrated into the inner and outer layers of the coat monitor the outer temperature near the fire-fighter and on the inside of the coat close to the body. These are attached to two LED displays, one on the sleeve and the other on the back. The LED display on the upper left shoulder is visible to other fire-fighters of the team and indicates potentially critical situations. The LED display on the lower sleeve indicates dangerous heat levels both inside and outside the turnout gear. A small box in the inner liner of the coat contains a battery and a control chip that calculates the temperature and activates the LED displays. The sensors are covered in flexible waterproof plastic to protect against fluids. The microelectronics is durable and can withstand at least 25 wash cycles. The only thing needed before cleaning is to remove the battery from a pocket in the liner. No maintenance is required—only the battery needs to be changed.

Power generating clothing. The smart clothing can harvest the body power of its wearer by harnessing the static electricity generated from regular body movement. Currently, this electrical energy is wasted. The vision is to collect this waste energy by weaving conductive filaments into the fabric. Everyday clothing can thus be transformed into a power-generating source.

GPS jacket that knows where it is. Wearable jackets integrated with a hands-free/Bluetooth telephone and iPod are making their way into retail stores. Interactive Wear's expertise lies in determining how to best split up the individual electronic components used in jackets, vests or backpacks that offer such capabilities, miniaturising these components as necessary, optimising power input, and interlinking all these elements using a network of signaling and power-supply lines that are integrated into the fabric.

The advantages of this type of integration for numerous application scenarios are obvious. The overall

system integrated into the garment is miniaturised and up to 30 per cent lighter than standalone solutions. It can be worn comfortably and unpretentiously, and is easy to operate, protected and practically 'impossible to lose.'

Besides a cellular phone, an MP3 player, headphones, a microphone and a water-tight and impact-resistant sleeve keypad as well as an emergency call button have all been integrated into a terminal that is referred to as a 'GPS eye.' This terminal is used for determining and transmitting positional data in order to locate the people. In this way, it is not only possible to determine the location of

gration of all the individual electronic components and systems for augmented reality into a vest. This wearable computing integration provides new degrees of freedom.

Other conceivable areas of application include industrial production, medical device technology (augmented reality operations) and nearly all areas of research and development.

Challenges to overcome

Efforts are aimed at staying close to conventional large-scale cutting and sewing techniques for making electronic textile clothing.

But, cutting an electronic cloth would make it more difficult to make

Carbon nanotubes are regarded as one of the most versatile nanomaterials available because of their mechanical strength and electrical properties

the jacket's wearer with GPS accuracy but it is also possible for GPSoverIP to even enable real-time determination of the person's location within a building. Compared to conventional solutions that are not integrated into garments, integration of the active GPS antenna into the epaulette on the jacket's shoulder ensures optimal reception of GPS signals.

Areas of application for the Know Where solution are many, ranging from luxury, entertainment and leisure (mountaineering, sailing, etc) to safety, traffic and logistics.

Augmented reality. Augmented reality (AR) is not new, particularly in the automotive industry, where this technology is used to link the virtual world with the real one. It can be used, for example, to guide a mechanic through a repair procedure. When wearing a pair of head-mounted display (HMD) glasses, mechanics can be provided with additional three-dimensional information, which enables them to see beyond the real environment to view virtually animated components, tools or instructions.

The new technology here is inte-

good connections between different parts of the garment. One solution to this problem could be the manufacturing of seamless clothing, which would avoid the need for cutting and stitching altogether. The cost of developing and manufacturing such sophisticated fabrics is likely to put them beyond the reach of the fashion industry for the time being. So connections will be the main point of weakness in electronic clothing.

Moreover, researchers are yet to answer the million-dollar question—perhaps critical to consumer acceptance—about the washing of electronic fabrics.

Another challenge is to ensure users' security and privacy from hackers.

However, despite the technical obstacles, researchers involved in the development of interactive electronic clothing appear universally confident that smart electronic clothing envisaged for different applications to the wearer is only a matter of time! ●

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